MECHANICAL ENGINEERING PROJECT PORTFOLIO

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ABOUT ME!

I'm a Mechanical Engineer with hands-on experience across design, manufacturing, and process optimization. I graduated from Texas A&M University in May 2025 with a B.S. in Mechanical Engineering, where I worked in the INVENT Lab researching tribology for haptic perception in tactile robotics.

Professionally, I've contributed to product design and manufacturing at Ingersoll Rand, process improvement at Tesla's Fremont factory, and now steel production and optimization at Nucor Steel. My work spans CAD design, system analysis, and data-driven process control – all driven by a focus on building reliable, efficient, and scalable mechanical systems.

I'm motivated by solving real engineering problems and finding practical ways to make machines, materials, and processes perform better every day.

LEAK PROTECTION SYSTEM FOR SERVER RACKS

CONTEXT AND OBJECTIVE

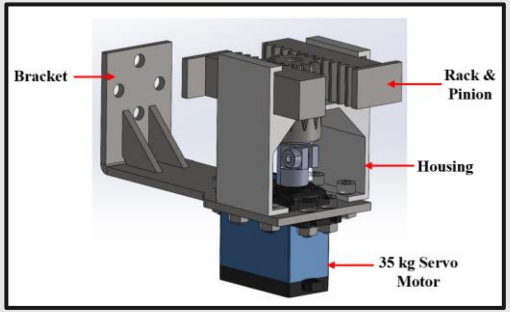
As AI pushes computers to new performance heights, heat is skyrocketing. Water cooling can handle it—but even a small leak in a server rack can destroy millions in equipment and halt operations. My capstone project at Texas A&M, in partnership with Dell, tackles this challenge: **mitigating the risk of leaks.**

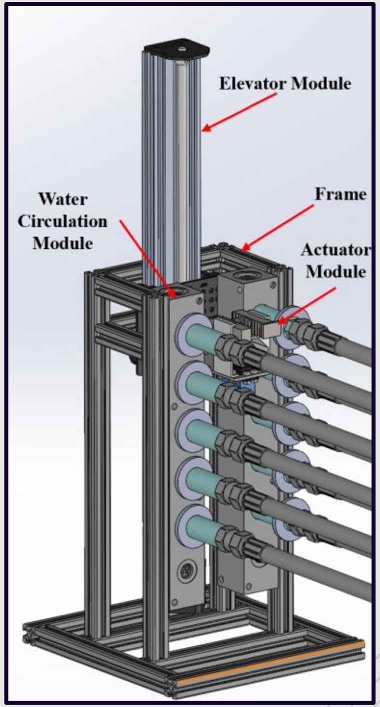
SOLUTION

I developed an automated system that **isolates and stops the flow of liquid to the specific server** where a leak is detected. This prevents the need to shut down the entire rack, allowing the leak to be addressed before it causes significant damage.

KEY FEATURES

This **patent-pending** system offers a highly effective solution for managing server leaks. It is **cost-effective** and **easily scalable** across data centers with hundreds of racks, making it practical for large-scale deployment. The system can **stop leaks within 15 seconds** of detection





TESLA FREMONT:DOOR-TRIM CONVEYOR

CONTEXT AND OBJECTIVE

A new Tesla Model 3 was in development with six variants of the 'door-trim' component, enabling the launch of an **AGV-driven process**. Since forklifts could not load racks directly onto the AGVs, an **interface system** was to be developed to facilitate the transfer.

SOLUTION

I developed an **automated conveyor system** that received racks from forklifts and transferred them to the AGVs. The system incorporated **pneumatic circuits** to operate stop cylinders, **electronic components** to acknowledge rack receipt and delivery, and **intelligent logic** to track the number of racks held, alerting operators in case of shortages.

KEY FEATURES

After successfully validating the system, it was **replicated six more times** across different areas of the factory, establishing itself as the **standard forklift-AGV interface**. The system also allows operators to manually unload racks in the event of an automated component failure, ensuring **uninterrupted operation**.

ELECTROADHESION SETUP

CONTEXT AND OBJECTIVE

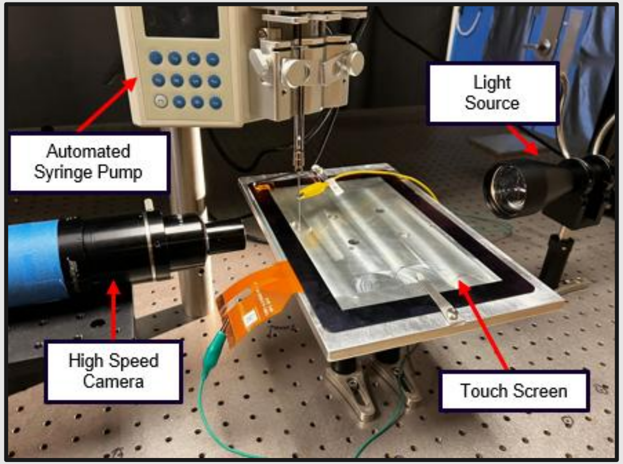
When using **electro adhesive touch screens**, researchers noticed that some parts of the screen get oily faster than others. To understand this phenomenon, I was tasked to build a setup to capture how tiny drops of finger oil move and change when electricity is applied.

SOLUTION

I developed **a high-speed camera system** to capture the behavior of oil droplets under an electric field. The setup allowed **precise adjustment in all three directions** to maintain perfect focus, and an automated syringe system delivered oil droplets onto the charged screen in a controlled and repeatable manner.

KEY FEATURES

This system enabled us to capture high-resolution images of droplets under varying currents, revealing that higher voltages lead to stronger capillary bridge formation. These findings contributed to a **published study** in ***Advanced Materials Technologies.***



https://advanced.onlinelibrary.wiley.com/doi/full/10.1002/admt.202300213

OPTICAL COHERENCE TOMOGRAPHY SETUP

CONTEXT AND OBJECTIVE

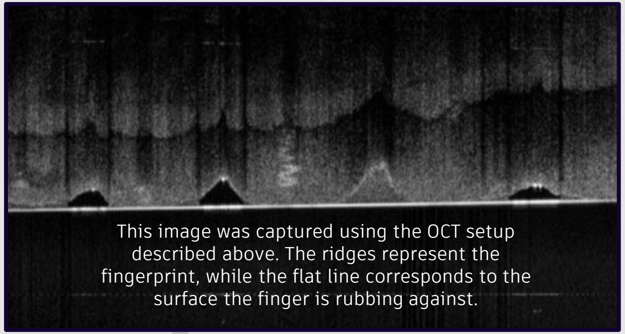
At INVENT Lab, we explored ways to add **tactile feedback to robotic end-effectors** by studying how human fingertips sense different surfaces. Using **Optical Coherence Tomography**, my task was to develop a setup to image **fingertip interactions** as various surfaces moved across them.

SOLUTION

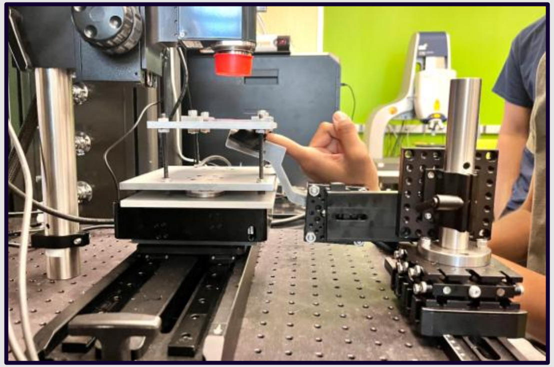
A complete **electromechanical setup** was designed, fabricated, and programmed to image fingertip interactions and measure the normal force exerted as surfaces oscillated over the finger. The system integrated **load cells, DAQ cards, and an OCT unit**, all engineered to ensure the finger remained comfortable and free from stress during testing.

KEY FEATURES

The system successfully captured **high-resolution images** of subsurface skin layers and is now being used to gather data for **publications in leading tribology research journals**. Its versatility and precision have made it a core experimental setup for ongoing studies **on tactile perception and skin–surface interactions**.



This image was captured using the OCT setup described above. The ridges represent the fingerprint, while the flat line corresponds to the surface the finger is rubbing against.



THANK YOU!

Please feel free to connect with me!

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